SUPERSEDING
MIL-D-81893 (AS)
17 MARCH 1975

MILITARY SPECIFICATION

DISPENSING SET,

COUNTERMEASURES CHAFF AN/ALE-41

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

*1.1 Scope - The equipment covered by this specification shall dispense chaff capable of producing confusing signals in radar or other electronic weapons systems.

*1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Applicable Paragraph
Dispensing Set	AN/ALE-41	3.5.1
Control Indicator[1]	C-10004/ALE-41	3.5.2
Interconnect Cable	Part No. 112678-500(W9)	3.5.2
Interconnecting Box[1]	J-3386/ALE-41	3.5.3

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088 Wiring; Aircraft, Installation of

MIL-E-5400 Electronic Equipment, Aircraft, General Specification for

[1]The control indicator in conjunction with the interconnecting box shall provide a capability for in-flight selection of chaff deployment parameters. However, they shall not be required for the preset operational mode of the AN/ALE-41 system. (The Control Indicator and Interconnecting Box were previously designated Cockpit Control Indicator and Remote Control Unit respectively.)

COMMENTS OR RECOMMENDATIONS WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT MAY BE SENT TO THE PREPARING ACTIVITY WHEN A SELF-ADDRESSED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARS AT THE END OF THIS DOCUMENT, OR BY LETTER ADDRESSED TO THE COGNIZANT ACTIVITY.

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MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-A-8591	Airborne Stores and Associated Suspension Equipment; General Design Criteria for
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-L-19538	Lacquer, Acrylic-nitrocellulose Gloss (For Aircraft Use)

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionic Equipment and Systems, General Requirements for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	Interference Control Requirements, Aircraft Equipment
MIL-STD-470	Maintainability Program Requirements
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution

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MIL-STD-785 Reliability Program for Systems and Equipment Development and Production

MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of

MIL-STD-810 Environmental Test Methods

MIL-STD-2074 Failure Classification for Reliability Testing (Supercedes AR-34)

MS25212 Control Panel, Console Type Aircraft Equipment, Basic Dimensions

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, PA 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provisions for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400 except when the use of such parts would conflict with interchangeability requirements for previously produced models of this equipment. Component parts shall be selected in the following order of precedence:

(a) TX, ER or other higher reliability parts.

(b) Military standard parts.

(c) Military Specification parts (not listed in book form standards).

(d) Nonstandard parts previously approved for use in other designs.

(e) Nonstandard parts.

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(3) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10. Mechanical subassemblies shall be modularized to the extent practical to facilitate production assembly and rework and to facilitate changes. The use of cements and welds or other forms of permanent bonds which will consume appreciable rework time shall be avoided in the design of repairable assemblies. Assemblies that require machining, lapping, or custom fitting shall be avoided. Where two units must be matched and used with each other, an assembly drawing shall be created that identifies this requirement. When adjustment or alignments must be made, components shall be pinned, slotted, or scribed to permit rapid reassembly with minimum gauging and trial and error. Replacement of modules shall not require use of solder, cement or welding. The design shall be such that reverse or improper assembly, mating or installation is impossible. Hard wired/chassis mounted components are undesirable and will not be used without specific written approval of the procuring activity. Where possible, circuitry loops shall be contained on one shop replaceable assembly (SRA). In cases that the number of discrete components in a loop exceed the practical or physical space available on one SRA, mounting and packaging shall be in a manner most adaptable to maintainability. Nonrepairable or disposable-on-failure subassemblies may be used and are preferred, providing such subassemblies are sufficiently reliable and inexpensive to justify that concept, or where significant weight savings may be achieved, or where mechanical integration with or modification of existing design required encapsulated subassemblies, or in the case of microwave energy devices. Items recommended as nonrepairable or disposable-on-failure candidates shall be identified.

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3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

*3.3.1 Total Weight - The total weight of the equipment excluding cables shall be a minimum consistent with good design and shall not exceed 220 pounds empty. When fully loaded, the total gross weight may vary from 500 to 550 pounds maximum depending on chaff payload configuration or mix. Empty and full weight shall be clearly marked on Center Body Shell.

*3.3.2 Reliability - The contractor shall conduct a reliability program in accordance with an approved AN/ALE-41 Reliability/Maintainability Program Plan of paragraph 3.3.2.1. On a reorder from a supplier who has previously produced the equipment, the program plan previously used may be continued unless otherwise indicated in the contract or order.

*3.3.2.1 Reliability/Maintainability Program Plan - A plan for the organization, implementation and conduct of a coordinated AN/ALE-41 Reliability and Maintainability Program shall be prepared by the contractor and submitted to the procuring activity for review and approval. AN/ALE-41 Maintenance Plan, AVMP-116, dated 21 Nov 1978, MIL-STD-785, MIL-STD-470 and AR-10 shall be used for the preparation of the program plan.

*3.3.2.2 Operational Stability - The equipment shall be capable of deploying a full chaff load in either Continuous or Pulse Mode operation, while maintaining the selected dispense rate. pulse lengths, and intervals within specified tolerances, for at least 90% of the pod in-flight dispense cycles. An unsuccessful in-flight dispenser cycle is defined as the inability of a part of the AN/ALE-41 system to perform within specified limits, resulting in less than 80% of the chaff in 2 or more chaff rolls being dispensed, or which requires corrective maintenance action other than pod clean-up and reloading to ready the pod for the next mission.

3.3.2.3 Operating Life - The equipment shall have total operating life of 500 cycles with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

*3.3.2.4 Reliability in Mean Cycles Between Failure (MCBF) - The equipment shall have 50 mean dispensing cycles between failures when tested and accepted utilizing simulated chaff loads as outlined under the requirements of 4.4.3 (a dispensing cycle shall be defined as the dispensing, or simulated dispensing, of one complete chaff load).

*3.3.2.5 Time Totalizing Meter - The equipment shall contain a time totalizing meter in accordance with Specification MIL-M-7793.

Unit	Type of Meter
Panel Assembly	MS17321-2

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3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400 except when the use of such parts would conflict with the interchangeability requirements for previously produced models of this equipment.

*3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088 and MIL-W-16878.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400. Interchangeability to the piece part-level shall exist without regard to the source of manufacture or supply. In the event of conflict between interchangeability with previously produced models of this equipment and other requirements of MIL-E-5400, the interchangeability requirements shall take precedence.

*3.3.5.1 Nose/Tail Cone Interchangeability - The dispenser's Nose Cones shall be interchangeable without sacrificing snugness of fit and without adding requirement for tools for prying, pushing, pulling, etc. This same requirement applies to dispenser Tail Cones.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461 for Class ID equipment.

*3.3.7 Maintainability - The contractor shall conduct a maintainability program in accordance with an approved Reliability/Maintainability Program Plan of paragraph 3.3.2.1. On a reorder from a supplier who has previously produced the equipment, the program plan previously used may be continued unless otherwise indicated in the contract or order.

*3.3.7.1 General - The design, configuration, installation and operation of the equipment shall minimize maintenance/support requirements, including the need for special tools and support equipment, personnel skills, manpower, elapsed maintenance time and total maintenance time at all levels of maintenance. The design shall permit accomplishment of the maximum number of tasks at intermediate level.

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*3.3.7.2 MTTR (Mean Time to Repair) - At the intermediate maintenance level, it shall be possible to fault isolate a failed weapons replaceable assembly (WRA) to the faulty shop replaceable assembly (SRA) within the constraints specified below:

a. Mean Time to Fault Isolate = 15 minutes: the contractor's design shall allow fault isolation of the SRA using appropriate technical manual and test equipment, within 15 minutes. The 15 minutes shall not include "hook-up" or "warm-up" time. This fault isolation to the failed SRA shall be accomplished without ambiguity. Mean Time to Fault Isolate, MTFI, is given by the following expression:

$$MTFI = (\Sigma \lambda_i t_i)/(\Sigma \lambda_i)$$

where λ_i is the failure rate of the "ith" SRA, and t_i is the time required to accomplish fault isolation of the "ith" SRA.

b. The contractor design shall be such that with nonautomatic ground support equipment a failed SRA can be located without ambiguity within 45 minutes.

c. The AN/ALE-41 shall contain the following WRA's and SRA's. Each SRA is a subsystem/component of a WRA:

Weapons Replaceable Assembly (WRA)	Shop Replaceable Assembly (SRA)
Nose Cone Assembly	Nose Cone Static Nostril Adapter Variable Orifice
Center Pod Assembly	Spider Component Assembly Power Amplifier Assembly Gear Box Drive Assembly Drive Motor Potentiometer Gear Box Assembly End of Cycle Control Assembly Tachometer Connector Assembly Cover Assembly Frame and Panel Assembly Amplifier Circuit Card Pulse Generator Circuit Card Panel Assembly
Air Shield and Converter Assembly	Clutch Assembly Converter
Cartridge Assembly	Frame Assembly Guide Roller Assembly Takeup Roller Assembly
Tail Cone Assembly	
Control Indicator (C-10004/ALE-41)	Circuit Card Control Unit Frame Assembly
Interconnecting Box (J-3386/ALE-41)	Circuit Card

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*3.3.7.3 Connectors - The design of connectors shall be such that interchanges, reverse or improper assembly, mating or installation is impossible. Screws on cable strain reliefs of connectors shall not protrude and interfere with (1) installation of adjacent components and (2) assembly of nose and tail cone to center pod component.

3.3.7.4 Cable and Wire Runs - Cable and wire runs shall be routed in such a manner as to avoid sharp bends, especially near connectors, and pinching. Protection shall be provided for cables routed through holes. Adequate clamping/support provisions shall be made. Easily removable replacement sections of cables shall be used where appropriate.

3.3.7.5 Encapsulation - Repairable SRAs/Assemblies shall not be encapsulated, potted or embedded. When a conformal coating is used, it shall be a transparent type which is easy to remove and reapply.

3.3.7.6 Fasteners - The different types of fasteners (screws, bolts, quick release fasteners, etc.) used in the system shall be minimized to reduce tool requirement for maintenance.

3.3.7.7 Remove/Replace Provisions - Design of the system shall make provisions for rapid removal and replacement of each WRA in the system and minimize requirements to remove or displace other WRAs/equipments. The design of the WRA shall continue this philosophy in that provisions shall be made for the rapid removal and replacement of SRAs and other replaceable items without removal of other parts, subassemblies, etc.

3.3.7.8 Standard, Special Tools and Test Equipment - The supplier's equipment shall be designed so that standard tools and test equipment or Ground Support Equipment (GSE) used with predecessor equipments may be used to the greatest possible extent to accomplish all necessary adjustments, maintenance and testing. Maximum utilization of existing GSE shall be made and special support equipment shall be developed only if existing equipment is impractical or will not accomplish the test function. The design of such equipment must be approved by the procuring agency.

3.3.8 Nomenclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

*3.3.8.1 Nameplate Attachment - The Nameplate shall be secured to the skin of the Center Body Shell by the use of rivets. The nameplate shall be installed starboard center adjacent to and below the hardback.

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*3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions for making laboratory bench tests:

Temperature	Room ambient ($25^{o} \pm 10^{o}C$)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 5.0 VAC, 400 Hz, 3 phase, 5A/phase
	28 ± 4 VDC, 1 AMP
	5 VAC/DC (Control Indicator lamps)

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment.

3.3.10.1 Acceleration - The equipment shall operate satisfactorily when subjected to the acceleration requirements of MIL-STD-810, Method 513.1, Procedures I and II.

*3.3.10.2 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of MIL-STD-810, Method 514.2, Procedures II B and II C, including Method 514.2 paragraph entitled, Summary.

*3.3.10.3 Humidity - The equipment shall operate satisfactorily when subjected to the humidity requirements of MIL-STD-810, Method 507.1, Procedure I, including Method 507.1 paragraph entitled, Summary.

*3.3.10.4 Shock - The equipment shall operate satisfactorily following subjection to the shock test of MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2, including Method 516.2 paragraph entitled, Summary. The shock test shall be conducted on a fully loaded pod.

*3.3.11 Warm-Up Time - The equipment shall operate properly under all specified service conditions with no warm-up.

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3.3.12 Input Electrical Power

*3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

(1) AC Power (Three Phase) 115/200V, 400Hz, Category B, 1280 VA

(2) DC Power 28 VDC, 1 AMP, Category B, 28 W.

3.3.12.2 Lighting Power - Input power for lighting shall require not more than 0.3 amp at 5 volts.

3.3.12.3 Degraded Performance - Degraded performance will be permitted for all voltage transients from 0 to 0.5 second duration as shown on Figure 2 of MIL-STD-704 for Normal Electric System Operation except performance shall not be affected by short pulses of less than 50 microseconds. Operation shall return to normal when the transient expires.

3.3.12.4 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage to the equipment or shall have suitable protective devices. The use of fuses and other protective devices shall be held to a minimum.

3.3.12.5 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.13 Cooling - The design for cooling the equipment shall comply with the requirements of Specification MIL-E-5400.

*3.4 Performance - Unless otherwise specified, the dispenser shall operate at the parameters given in Table 1, plus or minus a tolerance of 10 per cent, under both standard and extreme service input power conditions.

*3.4.1 Operation - The AN/ALE-41 shall be capable of simultaneously deploying chaff from 1 to 6 chaff rolls designed for use with this equipment. Ram air will enter the AN/ALE-41 dispenser through the removable nose cone and will enter the six air tubes where the chaff is injected. The specially designed chaff roll will consist of a mylar sandwich containing the chaff. In operation, the sheets of mylar will be separated onto two take-up rollers,

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allowing the chaff to be injected into the air tubes for ejection from the rear of the dispenser. The system shall be capable of either Continuous or Pulse Mode operation by presetting appropriate controls available on the Frame and Panel Assembly prior to flight, or by in-flight adjustment of the appropriate controls available on the Control Indicator, C-10004/ALE-41, provided the Interconnecting Box, J-3386, has been installed in the Frame and Panel Assembly. Chaff deployment selections during Continuous or Pulse Mode operation are given in Table 1.

TABLE 1

AN/ALE-41 CHAFF DEPLOYMENT CONTROL

CONTINUOUS MODE OPERATION

(CONTINUOUS/PULSE switch in CONTINUOUS position)

Controls Used	Selections
1. DISPENSE RATE Rotary switch, eleven selections. Chaff is dispensed from chaff rolls at the rate corresponding to the dispense rate selected.	OPEN (For alignment and Loading) 1.0 inch/second 2.0 2.5 3.0 3.5 4.0 5.0 6.0 7.0 8.0

PULSE MODE OPERATION

(CONTINUOUS/PULSE switch in PULSE position)

Controls Used	Selections
1. DISPENSE RATE	Same selections as in Continuous Mode
2. PULSE LENGTH Rotary switch, four selections. During each pulse cycle, the system is energized for a period of time corresponding to the pulse length selected.	0.5 seconds 1.0 2.0 4.0
3. PULSE INTERVAL Rotary switch, four selections During each pulse cycle, the system is de-energized for a period of time corresponding to the pulse interval selected.	0.5 seconds 1.0 1.5 4.0

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3.5 Detailed Requirements

*3.5.1 Dispensing Set, Countermeasures Chaff AN/ALE-41 - The Dispensing Set shall meet the following requirements:

*3.5.1.1 Function - The Dispensing Set shall provide the capability of dispensing chaff from compatible chaff rolls in a manner determined by (1) preselection of control settings on the self-contained Frame and Panel Assembly or (2) by in-flight selection of dispenser parameters utilizing the Control Indicator, provided the Interconnecting Box is installed in the Dispensing Set. Operating modes and available control settings shall be as given in Table 1.

*3.5.1.2 Form Factor and Mounting - The Dispensing Set shall be aerodynamically designed to provide low drag and shall be equipped with 14 inch and 30 inch mounting lugs for mounting on standard shackles per MIL-A-8591. The 14 inch lugs shall be the MK 6 MOD 0 type. The 30 inch lug shall be the MK 30 MOD 0 type. Dust plugs shall be provided for the unused lug holes. The pod shall be painted per MIL-L-19538 with FED-STD-595 Code 36492 Flat Gull Grey. The Center Body Shell shall be clearly marked FORWARD and AFT. The size of the unit shall be as shown in Figure 1 (page 33). Center of gravity shall be within limits set by MIL-A-8591. The pod shall be marked denoting centers of gravity under full and empty conditions in a easily viewable area between lug suspension points. Circumferential rib locations on the pod shall be marked as guidelines for positioning supports when transporting the AN/ALE-41.

3.5.1.3 Weight - See paragraph 3.3.1.

3.5.1.4 Input Electrical Power - See paragraph 3.3.12.1.

*3.5.1.5 Contents - The Dispensing Set AN/ALE-41 shall contain the following subassemblies and components:

Subassembly	Quantity
1. Center Pod Assembly	1
a. Air Shield and Converter Assembly	1
(1) Clutch Assembly	1
(2) Converter	1
b. Spider Component Assembly (Drive System)	1
(1) Gear Box Drive	1
(2) Drive Motor	1
(3) Potentiometer Gear Box Assembly	1
(4) Tachometer Connector Assembly	1
(5) Cover Assembly	1

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Subassembly	Quantity
c. Frame and Panel Assembly	1
(1) Amplifier Circuit Card	1
(2) Pulse Generator Circuit Card	1
d. Panel Assembly	1
(1) Power Amplifier	1
(2) End of Cycle Control	1
2. Nose Cone Assembly	1
a. Nose Cone	1
b. Static Nostril Adapter	1
c. Variable Orifice	1
3. Tail Cone Assembly	1
4. Cartridge Assembly	6
a. Frame Assembly	1/cartridge
b. Guide Roller Assembly	2/cartridge
c. Takeup Roller Assembly	2/cartridge

*3.5.1.5.1 Center Pod Assembly - The Center Pod Assembly shall be equipped with mounting lugs as described in paragraph 3.5.1.2 and shall contain the subassemblies listed above. The cartridge assemblies shall be loaded into the Center Pod Assembly (illustrated in Figure 1).

*3.5.1.5.1.1 Air Shield and Converter Assembly - The Air Shield and Converter Assembly is a major mechanical internal structure and shall provide security for the chaff rolls and drive assembly.

*3.5.1.5.1.1.1 Clutch Assembly - The Clutch Assembly shall interface with the spline on the number six chaff roll and shall provide stability to the drive train. The clutch shall also prevent the chaff rolls from counter rotating when the drive motor is not operating.

*3.5.1.5.1.1.2 Converter - The Converter shall be capable of converting 115 VAC, 400 Hz, 3 phase 5A/phase to 28VDC, 50 Amps to operate the dispenser drive system.

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*3.5.1.5.1.2 Spider Component Assembly (Drive System) - The Spider Component Assembly shall provide mounting provisions for the drive system. The drive system shall have the capability of simultaneously driving six chaff rolls, comprising a full chaff load, at the speeds specified in Table 1. It shall also be capable of Pulse Mode operation as specified in Table 1. All drive system components shall be readily accessible with the tail cone removed.

*3.5.1.5.1.2.1 Drive Motor - The Drive Motor shall be the prime mover of the drive system., It shall be capable of driving six full chaff rolls at a dispense rate of 8 inches per second throughout the dispense cycle.

*3.5.1.5.1.2.2 Gear Box Drive - The Gear Box Drive shall provide the interface between the Drive Motor and the drive system in the dispenser. It shall have a speed reduction ratio of 74.04:1.

*3.5.1.5.1.2.3 Potentiometer Gear Box Assembly - The Potentiometer Gear Box Assembly shall provide the interface, thru a flexible mechanical shaft, between the main drive sprocket and the End of Cycle Control. It shall provide a speed reduction of 50:1. (i.e., for every 50 revolutions of the main drive sprocket the flexible drive shall make 1 revolution).

*3.5.1.5.1.2.4 Tachometer Connector Assembly - The Tachometer Connector Assembly shall provide feedback voltage, proportional to the Drive Motor speed, to the Frame and Panel Assembly where it is used to maintain correct motor speed.

*3.5.1.5.1.2.5 Cover Assembly - The Cover Assembly shall provide the interface between the Frame and Panel Assembly and the dispenser. All the power and control signals shall be coupled from the Cover Assembly to the Frame and Panel Assembly thru a rack and panel connector located inside and at the back of the Cover Assembly. The Frame and Panel Assembly shall have a mating connector and shall slide into the Cover Assembly. The Frame and Panel Assemblies shall be interchangeable.

*3.5.1.5.1.3 Frame and Panel Assembly - The Frame and Panel Assembly shall contain the Amplifier and Pulse Generator circuits composed of two printed circuit boards and associated switches. The dispenser control switches shall be mounted on the front panel of the Frame and Panel Assembly and shall provide the following controls:

Nomenclature	Function
CONTINUOUS PULSE	Two position toggle switch shall select dispenser operating mode (reference Table 1).

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Nomenclature	Function
DISPENSE RATE	Rotary switch with stops shall control rate at which chaff is dispensed from chaff rolls in either Continuous or Pulse Mode operation (reference Table 1).
PULSE LENGTH	Rotary switch with stops shall control the dispenser "on" time for each pulse cycle during Pulse Mode operation (reference Table 1).
PULSE INTERVAL	Rotary switch with stops shall control the dispenser "off" time for each pulse cycle during Pulse Mode operation (reference Table 1).
LOCAL/REMOTE	Two-position toggle switch shall select whether dispenser control shall be accomplished by control selections on the front panel of the Frame and Panel Assembly (LOCAL position) or at the Control Indicator (REMOTE position).

*3.5.1.5.1.3.1 Amplifier Circuit - The Amplifier Circuit shall provide control for the Power Amplifier.

*3.5.1.5.1.3.2 Pulse Generator Circuit - The Pulse Generator Circuit shall provide the circuitry which controls the "on" and "off" times of the dispenser while operating in the Pulse Mode.

*3.5.1.5.1.4 Component Panel Assembly - The Component Panel Assembly shall provide a mounting surface and interconnections for the elapsed time meter, RF filter, main power contactor (relay), power terminal board, circuit breakers, End of Cycle Control, and Power Amplifier.

*3.5.1.5.1.4.1 Power Amplifier - The Power Amplifier shall amplify the motor drive current, as required, to maintain the motor speed selected on the Frame and Panel Assembly.

*3.5.1.5.1.4.2 End of Cycle Control - The End of Cycle Control shall be mechanically driven by a flexible drive from the Potentiometer Gear Box Assembly and shall provide the following functions:

(1) Automatically turn the dispenser off after a single cycle of operation.

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(2) Provide a visual indication of quantity of chaff remaining in the dispenser.

(3) Provide to the Control Indicator an electrical signal for remote indication of chaff remaining in the dispenser.

(4) Provide the reference drive voltage to the Frame and Panel Assembly. The reference drive voltage shall come from a voltage ratio potentiometer located in the End of Cycle Control box which makes one (1) revolution for every fifty (50) revolutions of the main drive sprocket. Input voltage is 15 VDC, the output reference voltage has a 1/x form increasing from near 0 to 15V.

*3.5.1.5.2 Nose Cone Assembly - The nose cone shall include a variable inlet orifice designed to automatically adjust to varied flight profiles so as to maintain internal pod pressure within limits required for proper chaff deployment. This profile shall encompass the range from Mach 0.3 to 1.2. A Static Nostril Adapter shall provide a means for sampling external air pressure. The nose cone shall be removable to permit chaff loading. The variable orifice nose cone shall be attached to the structural case by three stainless steel toggle action, flush mounted latches.

*3.5.1.5.3 Tail Cone Assembly - The tail cone shall be removable to permit access to drive system components and the Frame and Panel Assembly. The tail cone shall be attached to the pod structure by three stainless steel toggle action, flush mounted latches. The tail cone shall be designed to prevent the use of protruding air tubes as pod lift points.

3.5.1.5.4 Cartridge Assembly - The cartridge shall serve as an interface between the chaff roll and the dispenser. When loaded within the dispenser, the loaded cartridges shall inter-couple with each other and comprise part of the mechanical drive system for dispensing the chaff payload.

*3.5.1.5.4.1 Frame Assembly - The Frame Assembly shall hold the chaff roll payload. The chaff roll shall be mounted in bearings in the lower half of the Frame Assembly. The upper half shall lock the chaff roll into place. A threaded base on the frame interior of the lower cartridge frame and use of a winged nut with a shoulder stop shall be used to secure the upper cartridge rack frame to the lower cartridge frame.

*3.5.1.5.4.2 Guide Roller Assembly - The Guide Roller Assembly shall provide for separation of the two sheets of mylar on the chaff roller. It shall be mounted on the lower half of the Frame Assembly, located such that the mylar sheets shall be separated above the axial slot of its associated air tube.

*3.5.1.5.4.3 Takeup Roller Assembly - The Takeup Roller Assemblies shall provide a means for winding up the mylar from the chaff rollers as the chaff is dispensed. Two takeup rollers shall be on each cartridge and shall be driven by the main drive system. They shall assist in un-

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rolling the chaff roll by being driven slightly faster than the chaff roll. They shall have internal slip clutches to assure that the mylar is maintained taut whether the motor is on or off. In addition, the 2 take up rollers in each cartridge shall each have installed inside them, a torsional spring which must rotate approximately 200 degrees to reach the slip clutch normal torque range of 0.7 to 1.0 ft lbs. Roller wall shall be aluminum.

*3.5.1.6 Electrical Connections - Provisions shall be made for connections to external circuits as follows (also reference Figure 1):

Reference Designation	Receptacle Type	Function
J404	MS3122-E-14-19P	Power/Control
J403	MS3122-E-22-55S	Power/Control (EA-6A Aircraft)

Connectors J403 and J404 shall be wired in parallel to provide Control Indicator interface capabilities thru both connectors.

*3.5.1.6.1 Receptacle Covers - Metal connector covers are to be provided for the connector receptacles. The unused receptacle shall be covered to protect personnel from 115/200 VAC shock hazard. Restraining chains shall not be used.

*3.5.1.6.1.1 Receptacle Cover Stowage - Two dummy connectors matching J403 and J404 shall be mounted on the underside of the tail cone for stowage of the unused or spare receptacle cover.

*3.5.1.6.2 Hazard Warning - Near each of the above receptacles the words "DANGER 115/200 VAC HAZARD" shall be printed or noted by a decal, either of which shall utilize suitably large printing.

*3.5.1.6.3 Exposed Connections - All exposed electrical connections, such as terminal board terminations, etc., shall be painted/covered with a non-conducting sealant conforming to MIL-E-5400, to preclude personnel shock hazard and shorting of electrical circuits.

*3.5.2 Control Indicator C-10004/ALE-41

*3.5.2.1 Control Indicator - The Control Indicator shall meet the applicable requirements of Specification MIL-C-6781 for Type I control panels. The lighting of the plastic panel shall conform to MIL-P-7788, Class 1-R, Type IV with lamp voltages of 5 volts AC.

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*3.5.2.2 Function - The Control Indicator shall provide in-flight operation of the dispenser including selection of the following dispenser parameters: mode, dispense rate, pulse length, and pulse interval (reference Table 1). In addition, the Control Indicator shall:

(1) Interface with the AN/ALR-45 RHAW system.

(2) Provide a chaff remaining quantity indication.

(3) Provide an automatic cycle function that shall automatically repeat the selected Pulse Mode parameters for 1, 2, 4, 6 or 8 cycles.

(4) Permit operation of two dispensers independently or simultaneously.

*3.5.2.2.1 Interconnect Cable - Provides the capability of parallel operation with another Control Indicator to permit operation of four dispensers.

*3.5.2.3 Form Factor - The Control Indicator shall conform to Military Standard MS25212 and the dimensions shall not exceed 5.75 inches in width, 6.00 inches in height and shall not extend more than 5.00 inches behind the front panel, excluding connectors.

3.5.2.4 Weight - The maximum weight of the Control Indicator shall be 5.0 pounds.

3.5.2.5 Mounting - The Control Indicator shall be mounted in accordance with Specification MIL-C-6781.

*3.5.2.6 Input Electrical Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

(1) DC Power - 28 VDC, 1 AMP, 28 Watts

(2) AC Power (Three Phase) 115/200V, 400 Hz, Category B, 1280 VA.

3.5.2.6.1 Lighting Power - See paragraph 3.3.12.2.

*3.5.2.7 Controls - The Control Indicator shall include the following controls:

OFF/POWER - Two-position toggle switch shall control application of power to the control circuits.

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LEFT/RIGHT	- Two-position toggle switch shall select dispenser to be connected to Chaff Remaining indicator lights.
DECM AUTO/OFF	- Two-position toggle switch. DECM AUTO position shall permit system to be automatically activated by appropriate signals from the RHAW system.
PROG/OFF	- Two-position toggle switch shall be spring loaded to OFF position. PROG position shall initiate deployment of a programmed sequence of chaff pulses from both dispensers.
PULSE/PROG SET	- Rotary switch shall permit selection of number of chaff pulses to be deployed when the PROG/OFF switch is activated.
DISPENSER SELECT	- Rotary switch shall control application of power to dispensers and selects RIGHT, LEFT, or BOTH dispensers for operation.
PULSE/CONT/PI SET	- Three-position toggle switch. Two positions, PULSE/CONT, shall be used in the selection of the dispenser modes of operation. All three positions shall be utilized to change operational parameters; dispense rate, pulse length, and pulse interval.
DISPENSE RATE	- Rotary switch shall control rate at which chaff is dispensed from chaff rolls in either Continuous or Pulse Mode operation (Reference Table 1).
PULSE LENGTH	- Rotary switch shall control the dispenser "ON" time for each pulse cycle during Pulse Mode operation (Reference Table 1).
PULSE INTERVAL	- Rotary switch shall control "OFF" time for each pulse cycle during Pulse Mode operation (Reference Table 1).
CHAFF REMAINING LIGHTS	- Indicate quantity of chaff remaining in dispenser.
OPERATE LIGHTS	- Two lights, one for each dispenser. Lights shall illuminate when operating power is applied to the dispensers.

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*3.5.2.7.1 Switches - All switches shall be provided with mechanical stops to prevent inadvertent settings in blank positions.

*3.5.2.8 Electrical Connections - Four receptacles shall be provided on the Control Indicator for interconnecting to the dispensers and an additional Control Indicator as indicated below:

Reference Designation	Receptacle Type	Function
J1	MS3122-E-14-19P	Power inlet
J2	MS3122-E-22-55P	Outlet to left dispenser
J3	MS3122-E-22-55P	Outlet to right dispenser
J4	MS3122-E-14-19P	Power in/out for parallel operation of second Control Indicator.

*3.5.2.8.1 Connector Covers - Metal connector covers shall be provided for all four receptacles. They shall be connected to the Control Indicator cover by chains.

*3.5.3 Interconnecting Box J-3386/ALE-41

3.5.3.1 Function - The Interconnecting Box shall provide an interface between the Control Indicator and the Dispenser Set.

*3.5.3.2 Form Factor - The Interconnecting Box is approximately 4.50 inches in height, 5.12 inches in width and 1.70 inches in depth and shall be capable of being inserted in the Frame and Panel Assembly as a plug-in module.

3.5.3.3 Weight - The weight of the Interconnecting Box shall not exceed 2.0 pounds.

3.5.3.4 Contents - The Interconnecting Box shall contain the necessary circuitry to provide the interface between the Control Indicator and the dispenser.

*3.5.3.5 Electrical Connections - Connections to the Frame and Panel Assembly shall be as follows:

Reference Designation	Receptacle Type	Function
P1	M24308/4-5	Power/Signal input/output

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests
(5) Maintainability Demonstration Tests

4.2 Preproduction (First Article) Tests Preproduction tests shall be made on equipment representative of the production units to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of paragraph 4.7. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

*4.2.2 Scope of Test - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all of the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of specification MIL-T-5422 and MIL-STD-810, as required. In addition, the Dispensing Set shall be subjected to ground functional testing which shall demonstrate, to the satisfaction of the procuring activity, the capability of the dispensing set to dispense chaff in accordance with any possible combination of operating modes and control parameter selections listed in Table 1.

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(1) For the first half of this functional test, at least three different dispense rates shall be utilized in the continuous dispensing mode. Three different pulse length and pulse interval combinations, providing pulse mode duty cycles of 0.25, 0.50, and 0.89, with a dispense rate of 8.0 inches/second, shall be used for the last half of the test.

(2) The Control Indicator and Interconnecting Box shall be connected to the Dispensing Set and shall be utilized to control the dispenser.

(3) Satisfactory completion of this test requires that:

(a) During the performance of the test, chaff dispensing appears to be consistent with the operating modes and control selections.

(b) Post test inspection indicates that the dispenser is undamaged and free of malfunctions.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction samples shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the preproduction samples. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample(s) shall not be considered as one of the deliverable equipments under the contract.

4.2.4 Production Equipment - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

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4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

*4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all individual tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

*4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Acceptance tests shall be accomplished under the approved procedures of paragraph 4.7. Contractors not having adequate facilities for conducting all required tests shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish tests reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational accuracy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met, and to assure conformance with provisions of 3.5.1, 3.5.2, and 3.5.3 of the specification as appropriate.

4.4.1.2 Operational/Compatibility/Electrical Tests - Each equipment shall be subjected to test and inspections to assure mechanical and electrical compatibility with associated equipment. In addition, each equipment shall undergo electrical testing and be operated as required to check sufficient characteristics and record adequate data to assure satisfactory equipment operation. This testing shall be conducted in accordance with the approved Acceptance Test Procedures of paragraph 4.7 of this specification.

*4.4.2 Sampling Tests - Sampling tests shall be conducted in conformance with the provisions of MIL-STD-105.

*4.4.2.1 Units Subject to Sampling Tests - The following products, governed by this specification, shall be subject to sampling tests and each shall constitute a unit of product as defined by MIL-STD-105:

Dispensing Set AN/ALE-41
Control Indicator C-10004/ALE-41
Interconnecting Box J-3386/ALE-41

*4.4.2.2 Sampling Test Plan - Equipments shall be selected for sampling tests by the Government inspector in accordance with a sampling test plan which shall be prepared by the contractor and submitted to the procuring activity for review and approval. Unless otherwise specified by the procuring activity, the test plans shall be based on provisions for inspection level II and single sampling for normal inspection. (Tables I and II-A, MIL-STD-105), classification of defects, acceptable quality levels (AQL's), and lot size defined as follows:

*4.4.2.2.1 Classification of Defects - A defect is any nonconformance of the unit of product with specified requirements. A classification of defects is enumeration of possible defects according to their seriousness. With regard to the AN/ALE-41 Dispensing Set, defects may be classified as critical, major or minor as follows:

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Critical Defect: Any defect that is likely to result in failure to deploy chaff.

Major Defect: Any defect, other than critical, that is likely to materially reduce the system's chaff deployment capability.

Minor Defect: Any defect that is not likely to materially reduce the system's chaff deployment capability.

*4.4.2.2.2 Acceptable Quality Level (AQL) - The AQL is the maximum number of defects per hundred units that can be considered acceptable for sampling testing under this specification. Unless otherwise specified by the procuring activity, the AQL's for each class of defect shall be as follows:

Defect Class	AQL (Defects per hundred units)
Critical	0.01
Major	4.0
Minor	10.0

The right is reserved by the procuring activity to inspect every unit submitted by the contractor for critical defects and to reject the lot immediately when a critical defect is found.

*4.4.2.2.3 Lot for Sampling Tests - Unless otherwise specified by the procuring activity, sampling tests shall be conducted for each increment of production units scheduled for delivery in accordance with the contract schedule. The number of units in each increment shall constitute a lot for purposes of defining a sample size for testing in conformance with MIL-STD-105.

*4.4.2.3 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to examination of product and operational/compatibility/electrical tests as prescribed in 4.4.1.1 and 4.4.1.2 of this specification.

*4.4.2.4 Acceptance and Rejection - For any class of defects, if the number of defects found in the sample units subjected to testing is equal to or less than the acceptance number specified in the approved sampling test plan, the lot shall be considered acceptable. If the number of defects is equal to or greater than the rejection number, the lot shall be rejected. The procuring activity reserves the right to reject any lot if a sample drawn therefrom is found to contain one or more cricital defects.

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*4.4.2.4.1 Defective Units - The procuring activity reserves the right to reject any unit of product found defective during inspection whether that unit of product forms part of a sample or not, and whether the lot or batch as a whole is accepted or rejected. Rejected units may be repaired or corrected and resubmitted for inspection with the approval of, and in the manner specified by the responsible authority.

4.4.3 Reliability Assurance Test - Reliability Assurance Tests shall be conducted using MIL-STD-781 as a guide. Tests as required by both the Qualification Phase and the Reliability Acceptance Phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074.

*4.4.3.1 Reliability Qualification Phase Test - Reliability of the dispenser equipment shall be such that the Mean Cycles Between Failures (MCBF) shall be not less than fifty (50) dispensing cycles. (One dispensing cycle shall be defined as the dispensing of one complete load.) The qualification phase shall be 167 cycles in length. The test shall be conducted using two complete AN/ALE-41 systems. A minimum of forty-two (42) cycles will be required for one of the systems. The test units shall be subjected to a temperature cycle, in accordance with paragraph 5.2.3.1 of MIL-STD-781 with the following exceptions: (1) one-third of a simulated chaff load shall be dispensed starting immediately after the initiation of the heating period and remaining two-thirds of a simulated chaff load shall be dispensed following stabilization at the highest temperature; (2) in the event a dispensing cycle is longer than the temperature cycle, the dispensing of the simulated chaff load shall start immediately after the initiation of the heating period and be terminated on conclusion of the temperature cycle. The dispense rate, pulse length and pulse interval for the dispenser shall be changed after each temperature cycle so as to cover a representative portion of the operational spectrum of the AN/ALE-41 system. A minimum requirement shall be to operate at each dispense rate in Continuous Mode and to operate with all possible pulse length and pulse interval combinations at the minimum and maximum dispense rates in Pulse Mode. Additional Pulse Mode combinations may be specified when deemed necessary by the procuring activity. Test Level E of MIL-STD-781 shall be used. The equipment shall be vibrated the first ten (10) minutes out of each hour of the heating period. The Accept/Reject Criteria shall be defined as completion of the test with five (5) failures or less. The contractor shall record and report all failures. Failure analysis and corrective action to prevent re-occurrence of the failure shall be applied to each failure.

*4.4.3.2 Reliability Acceptance Phase Test - Each dispenser Spider Component Assembly including the Frame and Panel Assembly with Interconnecting Box J-3386/ALE-41 installed, and the Control Indicator C-10004/ALE-41 shall be tested for ten temperature cycles, conducted under the conditions of test level E of MIL-STD-781. Two complete dispensing cycles shall be conducted during each temperature cycle for a total of twenty dispenser cycles failure free. The drive motor shall be operated without load.

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Operating modes and control parameters shall be selected using controls available on the Control Indicator C-10004/ALE-41. Operating modes and control parameters shall be changed during testing to exercise a portion of the dispensing set control capabilities obtainable with the Control Indicator. A minimum requirement shall be to change operating parameters to provide at least six different dispensing cycles in Pulse Mode operation and four different dispensing cycles in Continuous Mode operation. The DECM and PROGRAM functions shall each be exercised at least once during testing. In the event a failure occurs, the equipment shall be repaired and the test repeated from the start. All failures shall be recorded and reported. Failure analysis and corrective action shall be applied to all failure occurrences in accordance with paragraph 4.4.5.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipment for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity).

*4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Upon approval of the procuring activity, test procedures previously approved for the preproduction tests may be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

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(3) Submit to the procuring activity, for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be 500 cycles duration and shall be conducted on equipments that have passed the individual test. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests). (Test Life accumulated during the Reliability Test may be counted toward the Life Test provided the entire 167 cycles is accumulated on a single equipment and equipment selected is in accordance with the table below).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

*4.5.1 Test Conditions - The Life Test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal Ground (0-5000 ft).
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency
D. C. Voltage	28 ± 4 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

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4.5.3 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subjected to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of cycles which the part operated prior to failure

*4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. The report shall include the data listed in paragraph 4.5.4 and the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such reports and proposals and determine whether they are acceptable.

4.6 Maintainability Demonstration

4.6.1 Maintainability Demonstration Test - The supplier shall demonstrate achievement of the requirements of paragraph 3.3.7 of this specification. The test procedures for Maintainability Demonstration shall include tests to utilize WRA and SRA test points for input signal application and output signal measurement, and for determination that the WRA or SRA is operating within its specified performance characteristics. These tests shall demonstrate, for example, WRA or SRA sensitivity, drive capability, gain, frequency response, or any other performance characteristics necessary to determine that the WRA or SRA is operating within its performance specification.

4.7 Test Procedures - The procedures used for conducting preproduction, acceptance and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract.

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Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.8 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance, and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall submit the equipment for acceptance.

4.9 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.10 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged and marked for the level of shipment specified in the contract, or order in accordance with Specification MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The equipment specified herein is intended to be used in land/carrier based and rotary wing aircraft. The equipment shall provide the capability of dispensing countermeasures chaff immediately upon activation by the pilot/operator. Any portion of the equipment which is exposed to the exterior of the hull of the carrying aircraft shall be able to withstand landing, taxi and take-off operations without damage to the equipment or to the aircraft.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

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(2) Selection of applicable levels of packaging and packing (see paragraph 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

*6.6 Definitions - The following definitions apply to this specification:

AN/ALE-41	This nomenclature refers to one complete dispenser including nose cone, tail cone, center pod, six cartridges, and dispensing mechanism.

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Cartridge	A mechanical device which serves as the interface between a Chaff Roll and the AN/ALE-41.
Chaff Roll	This term identifies the various chaff packaging configurations containing the payload for the AN/ALE-41 dispenser.
Control Indicator C-10004/ALE-41	This is an optional item of equipment designed for aircraft cockpit installation to provide a capability for in-flight selection and operation of AN/ALE-41 systems.
Cycle	One pod dispensing cycle shall be defined as the dispensing of one complete (simulated or actual) chaff roll load.
System	As used herein includes one AN/ALE-41 Dispensing Set plus six Chaff Rolls.
Pod	Pod is used to define the structural shell of the AN/ALE-41 and shall include the nose and tail cones.
Payload	The payload is the chaff carried in the chaff roll.
Interconnecting Box J-3386/ ALE-41	This is an optional item designed for installation within the dispenser to provide the interface circuits necessary for the dispenser control by the Control Indicator.

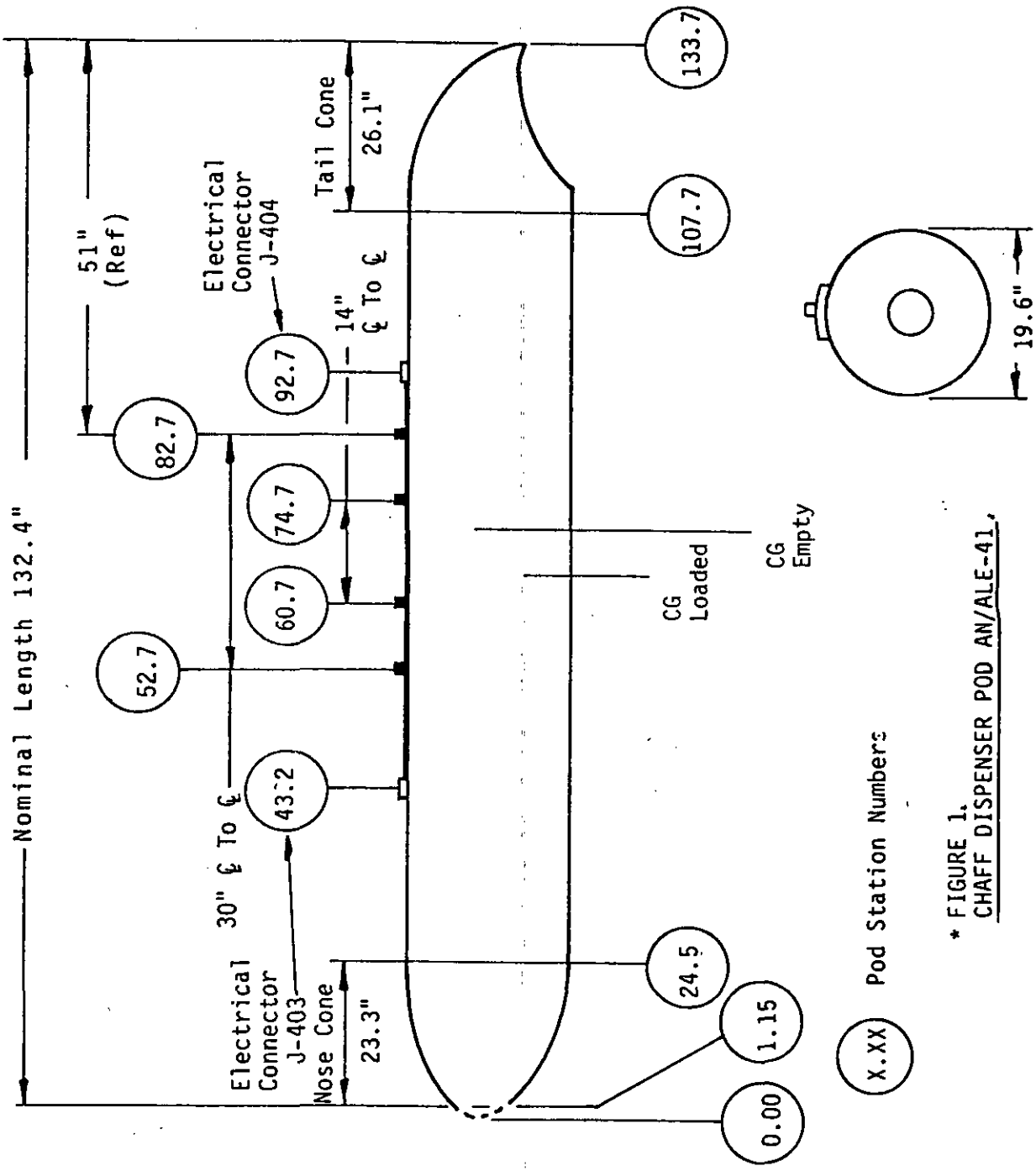
6.7 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 This specification is under the cognizance of Naval Air System Command AIR-53322M.

Preparing Activity:
NAVY-AS

Project No. 5865-N003

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* FIGURE 1. CHAFF DISPENSER POD AN/ALE-41.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE: MIL-D-81893A(AS) Dispensing Set, Countermeasures Chaff AN/ALE-41

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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